SI100B: Introduction to Information Science and Technology

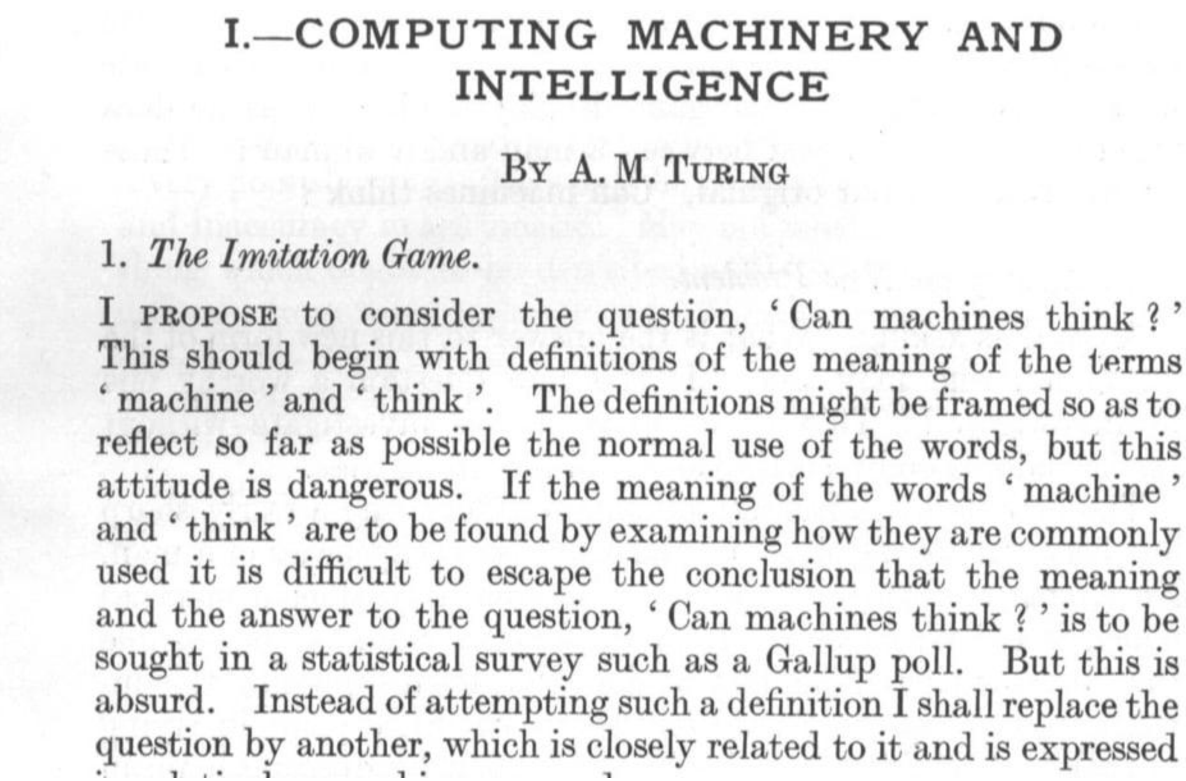
Lecture 13

By Kewei Tu of ShanghaiTech

A Brief Overview of Artificial Intelligence

A brief history

- Lots of early speculation & research
  - Turing: “Computing Machinery and Intelligence” (1950)

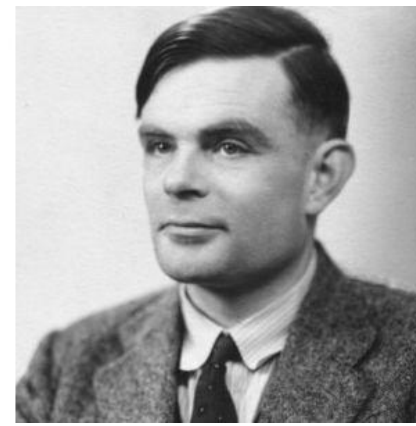


I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. *The Imitation Game.*

I propose to consider the question, ‘Can machines think ?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, ‘Can machines think ?’ is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed

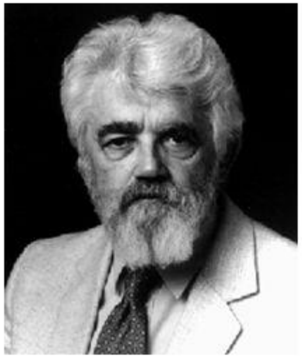


Alan Turing

A brief history

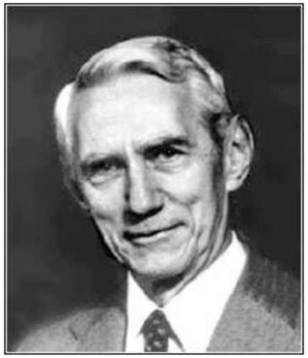
- Birth (1956)
  - Dartmouth workshop

Dartmouth Conference: The Founding Fathers of AI

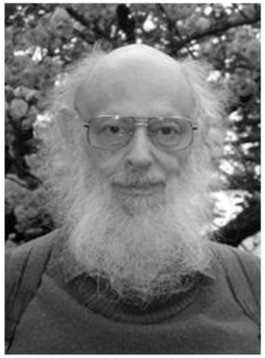


John McCarthy

Marvin Minsky



Claude Shannon



Ray Solomonoff

Alan Newell

Herbert Simon

Arthur Samuel

And several other people…

A brief history

- Great expectations (1950s-1960s)
  - A variety of methodology
    - e.g., symbolism, connectionism
- AI winter (1970s)
  - Downfall of perceptron
  - Lighthill report
- Boom (1980s)
  - Revival of neural networks
- More scientific methods (1990s-2000s)
  - Statistical approaches

Very rough timeline

A brief history

- 2010s
  - Rise of big data + big models + big computing (deep learning)
  - AI becomes one of the hottest areas
  - Great interest from industry and public
  - Many real-world applications
- The past few years
  - ChatGPT and the rise of large language (+X) models
  - Huge impact in academia, industry and general public

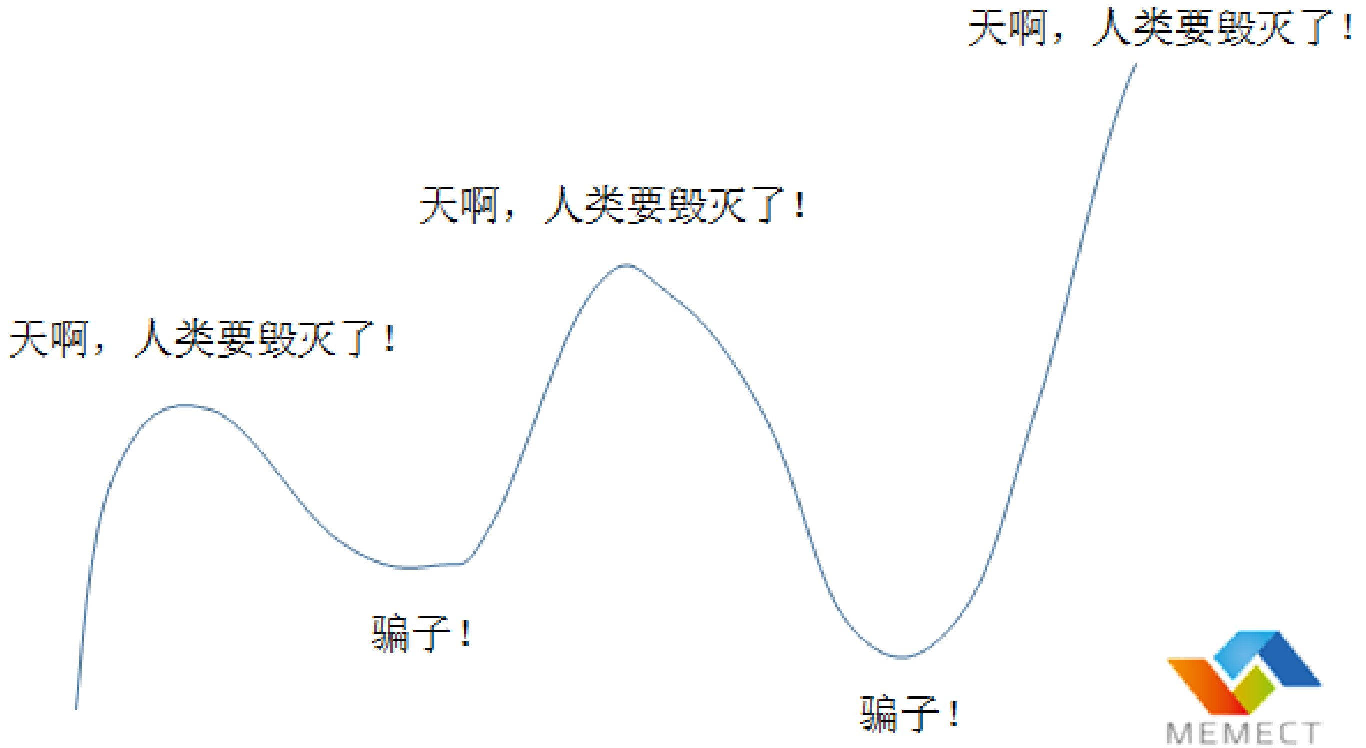
A NEW ERA

The Age of AI has begun

Artificial intelligence is as revolutionary as mobile phones and the Internet.

By Bill Gates | March 21, 2023 • 14 minute read

A brief history



Major Subfields of AI

Multi-Agent System

Integration

Robotics

Natural Language Processing

Modality-Specific

Computer Vision

Speech Recognition

Foundation

Machine Learning

Knowledge Representation & Reasoning

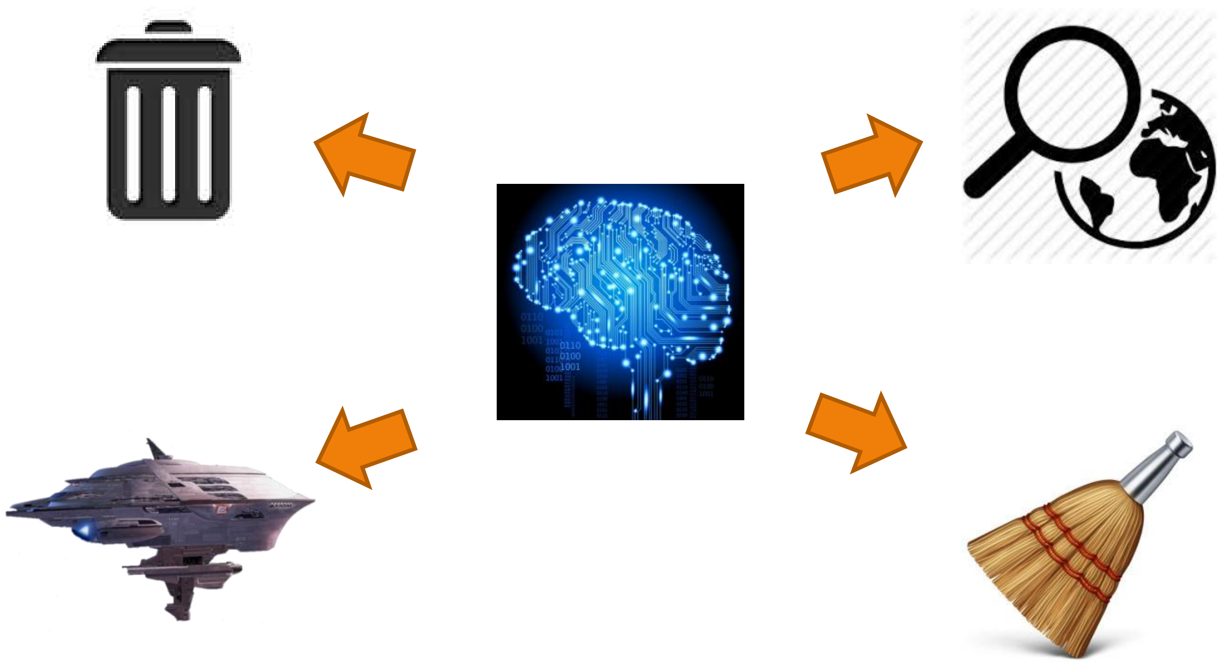
Uncertainty in AI

Strong AI vs. Weak AI

- Weak AI (Applied AI)
  - AI that accomplishes specific tasks

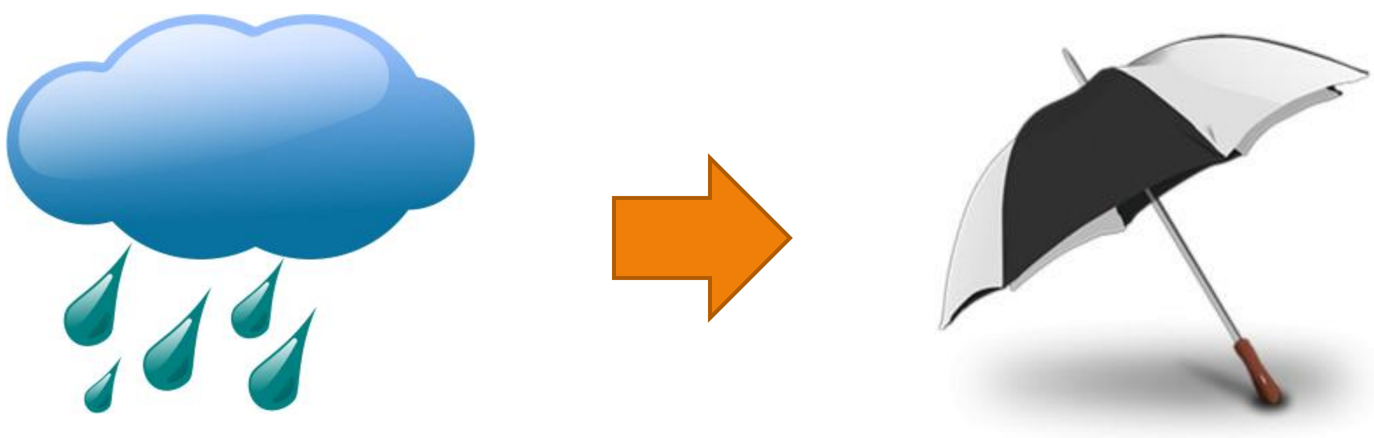
Strong AI vs. Weak AI

- Strong AI (General AI, AGI)
  - human-like intelligence – AI that could successfully perform any intellectual task that a human can



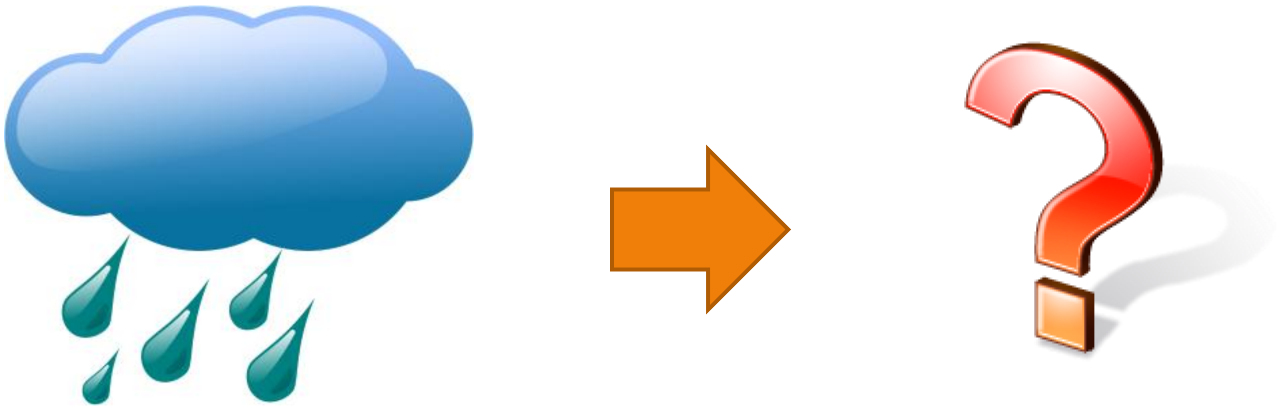
Central problems of (strong) AI

- Knowledge Representation (KR)
  - Knowledge: facts, beliefs, concepts, skills, … that are accumulated over time



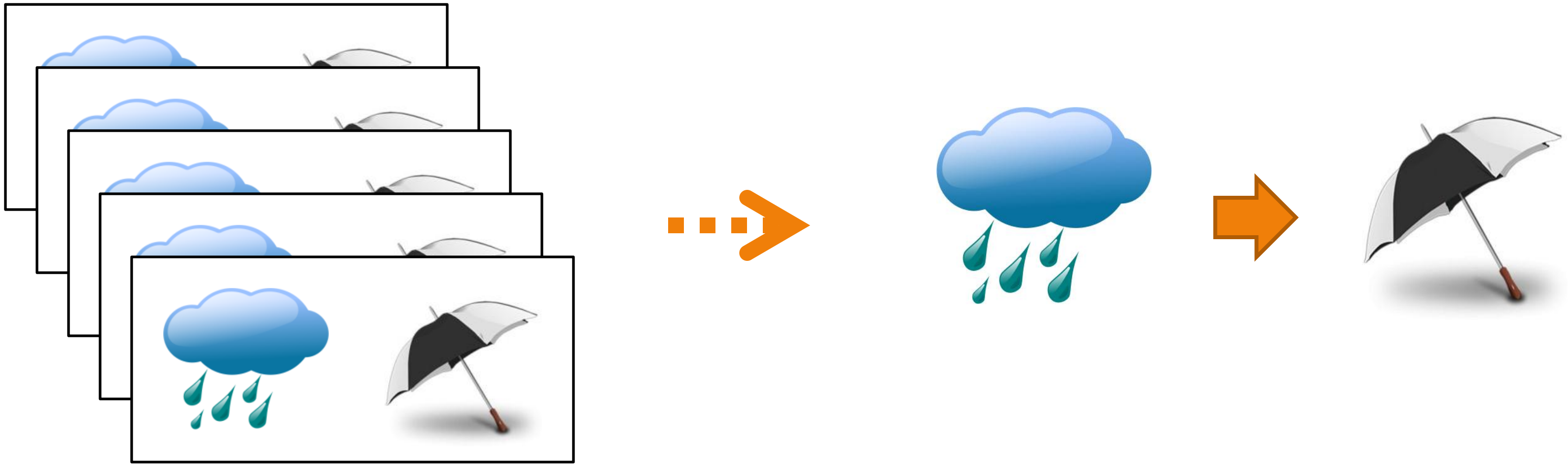
Central problems of (strong) AI

- Inference
  - How to utilize knowledge to derive new information based on existing information



Central problems of (strong) AI

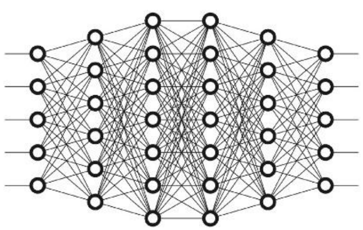
- Learning
  - How to accumulate knowledge from experience and education



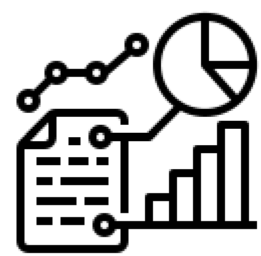
Three types of approaches

Symbolism

$$
\begin{array}{cccc}
+ & - & \times & \div \\
\neg & \vee & \perp & \cong \\
\in & \cap & \subseteq & \Sigma \\
\partial & \nabla & \wedge & \Pi
\end{array}
$$



Connectionism

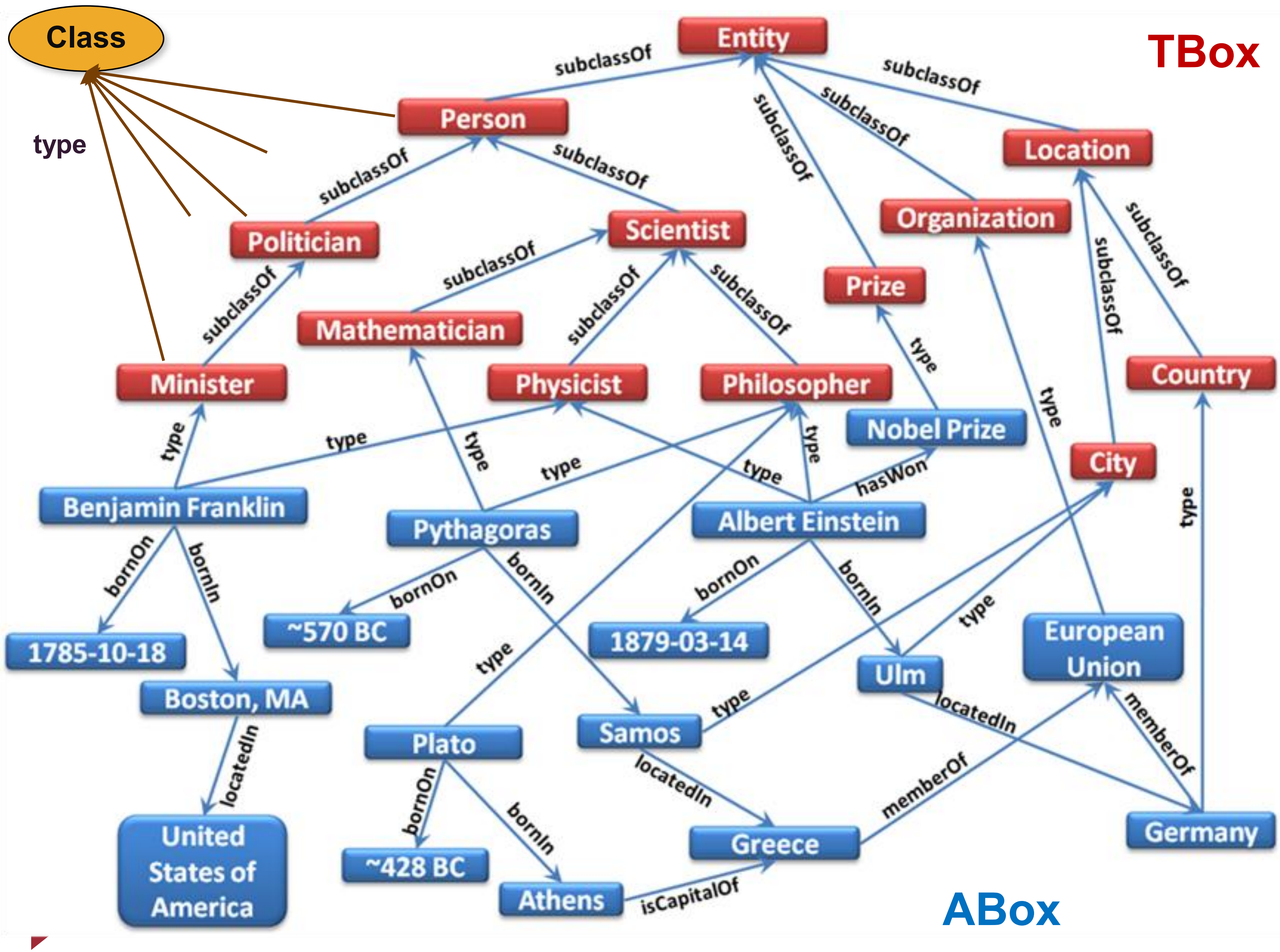


Statistical Approaches

Symbolism

- Representing knowledge with symbols and their compositions (expressions)

$$\forall x \forall y, Human(x) \wedge Place(y) \wedge At(x, y) \wedge Rain(y)$$
$$\rightarrow \exists z, Umbrella(z) \wedge Use(x, z)$$



Class
type
TBox
ABox
Entity
Person
Location
Organization
Politician
Scientist
Prize
Mathematician
Minister
Physicist
Philosopher
Country
City
Nobel Prize
Benjamin Franklin
Pythagoras
Albert Einstein
1785-10-18
~570 BC
1879-03-14
Ulm
European Union
Boston, MA
Samos
Plato
United States of America
~428 BC
Athens
Greece
Germany
subclassOf
type
bornOn
bornIn
hasWon
locatedIn
memberOf
isCapitalOf

Symbolism

- Inference and learning is done by manipulating symbols (e.g., logic)

Ex: Modus Ponens, a rule of inference

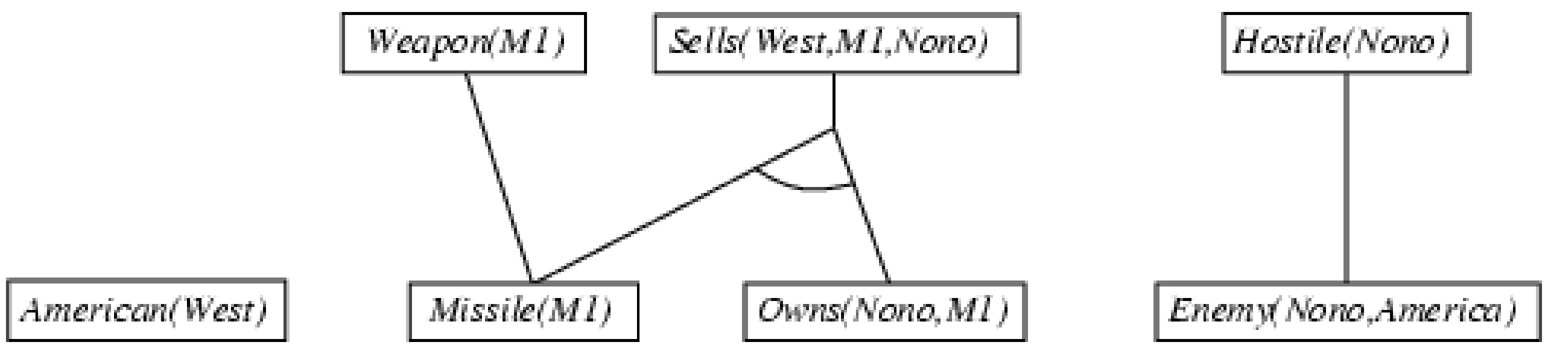
$$\frac{\alpha 1, \ldots, \alpha n, \quad \overbrace{\alpha 1 \wedge \ldots \wedge \alpha n}^{\text{premises}} \Rightarrow \overset{\text{conclusion}}{\beta}}{\beta}$$

First-order logic: forward chaining inference

- $American(x) \wedge Weapon(y) \wedge Sells(x,y,z) \wedge Hostile(z) \Rightarrow Criminal(x)$
- $Missile(x) \wedge Owns(Nono,x) \Rightarrow Sells(West,x,Nono)$
- $Missile(x) \Rightarrow Weapon(x)$
- $Enemy(x,America) \Rightarrow Hostile(x)$
- $Owns(Nono,M_1)$, $Missile(M_1)$, $American(West)$, $Enemy(Nono,America)$

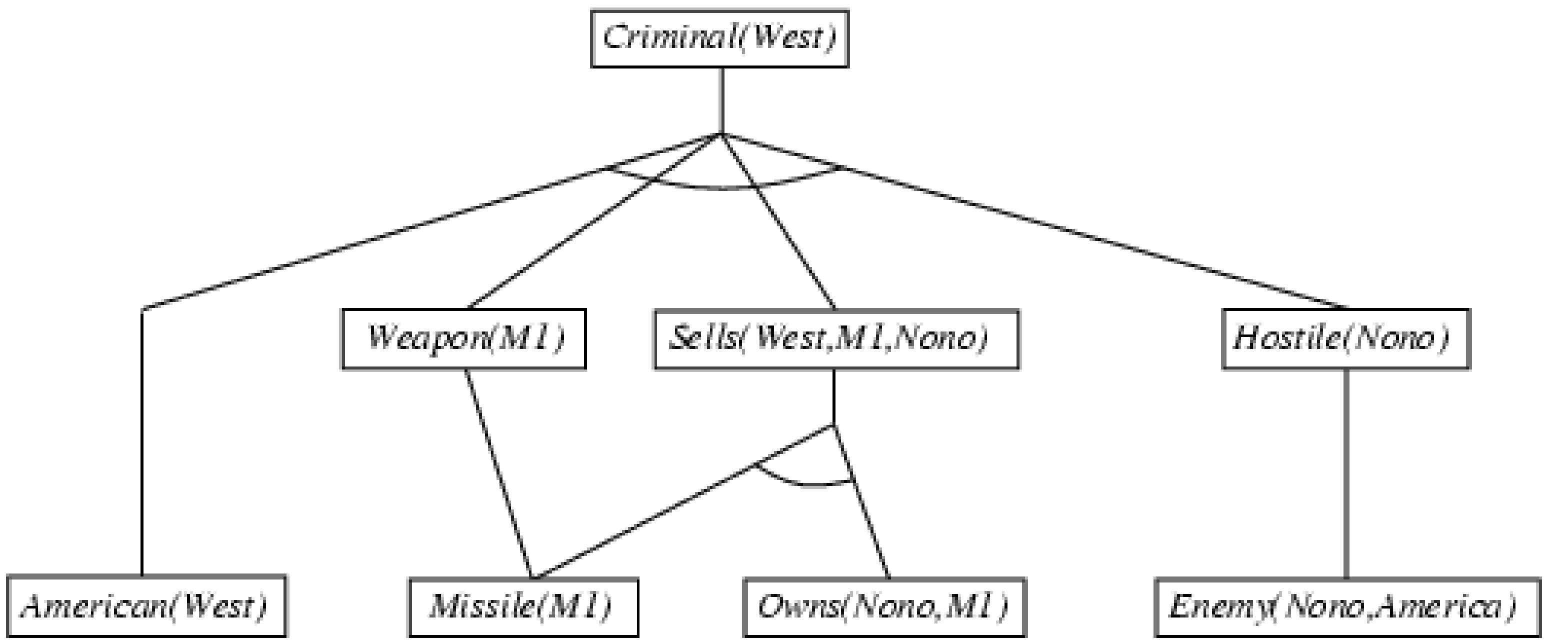
First-order logic: forward chaining inference

- *American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)*
- *Missile(x) ∧ Owns(Nono,x) ⇒ Sells(West,x,Nono)*
- *Missile(x) ⇒ Weapon(x)*
- *Enemy(x,America) ⇒ Hostile(x)*
- *Owns(Nono,M_1), Missile(M_1), American(West), Enemy(Nono,America)*



First-order logic: forward chaining inference

- *American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)*
- *Missile(x) ∧ Owns(Nono,x) ⇒ Sells(West,x,Nono)*
- *Missile(x) ⇒ Weapon(x)*
- *Enemy(x,America) ⇒ Hostile(x)*
- *Owns(Nono,M_1), Missile(M_1), American(West), Enemy(Nono,America)*

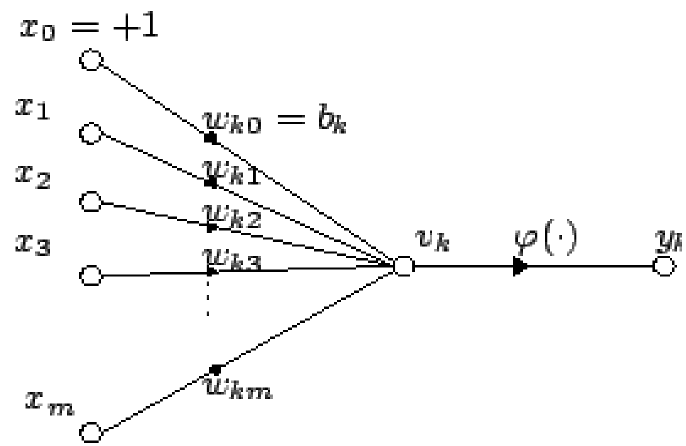


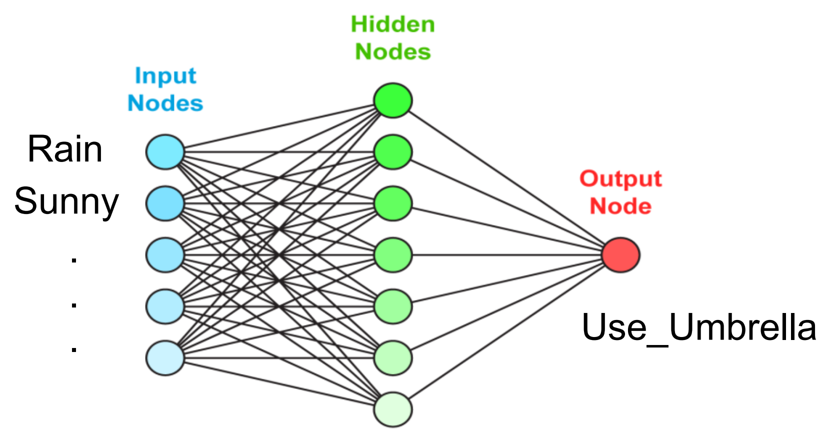
Symbolism

- History
  - Dominant during 1950s – 1980s
  - Fell out of favor in 1980s – 1990s
  - Integration with statistical approaches (2000s)
  - Integration with neural approaches (2010s)

Connectionism

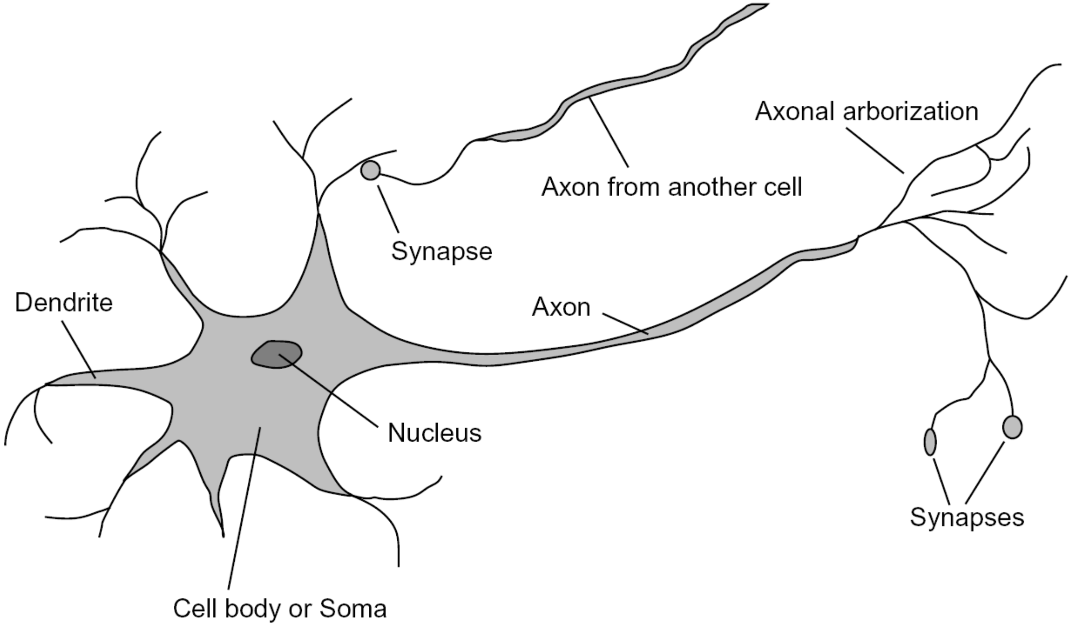
- Representing knowledge with interconnected networks of simple units
  - Neural networks
- Inference
  - Follow the computation specified by the network from input to output
- Learning
  - optimization of connection weights

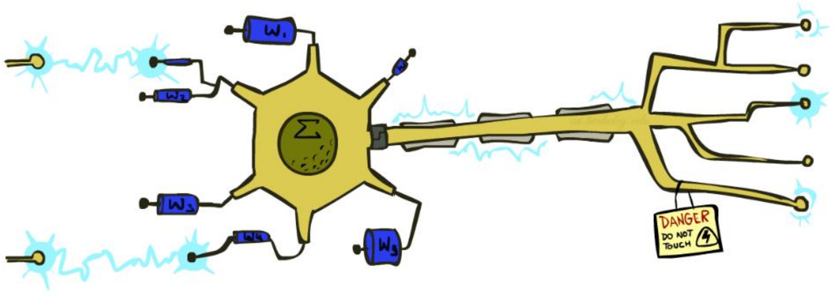




Neural networks: perceptron

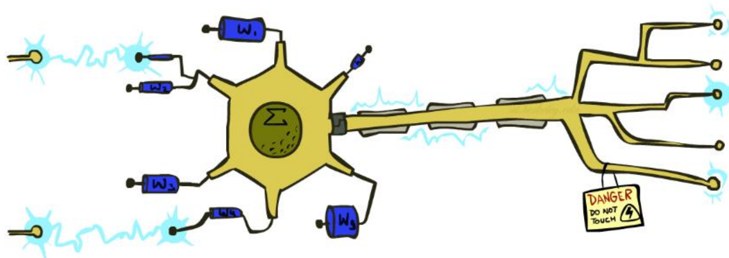
- Very loose inspiration: human neurons





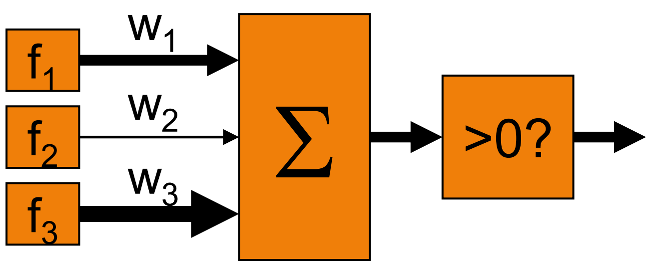
Neural networks: perceptron

- Inputs are feature values
- Each feature has a weight
- Computing weighted sum

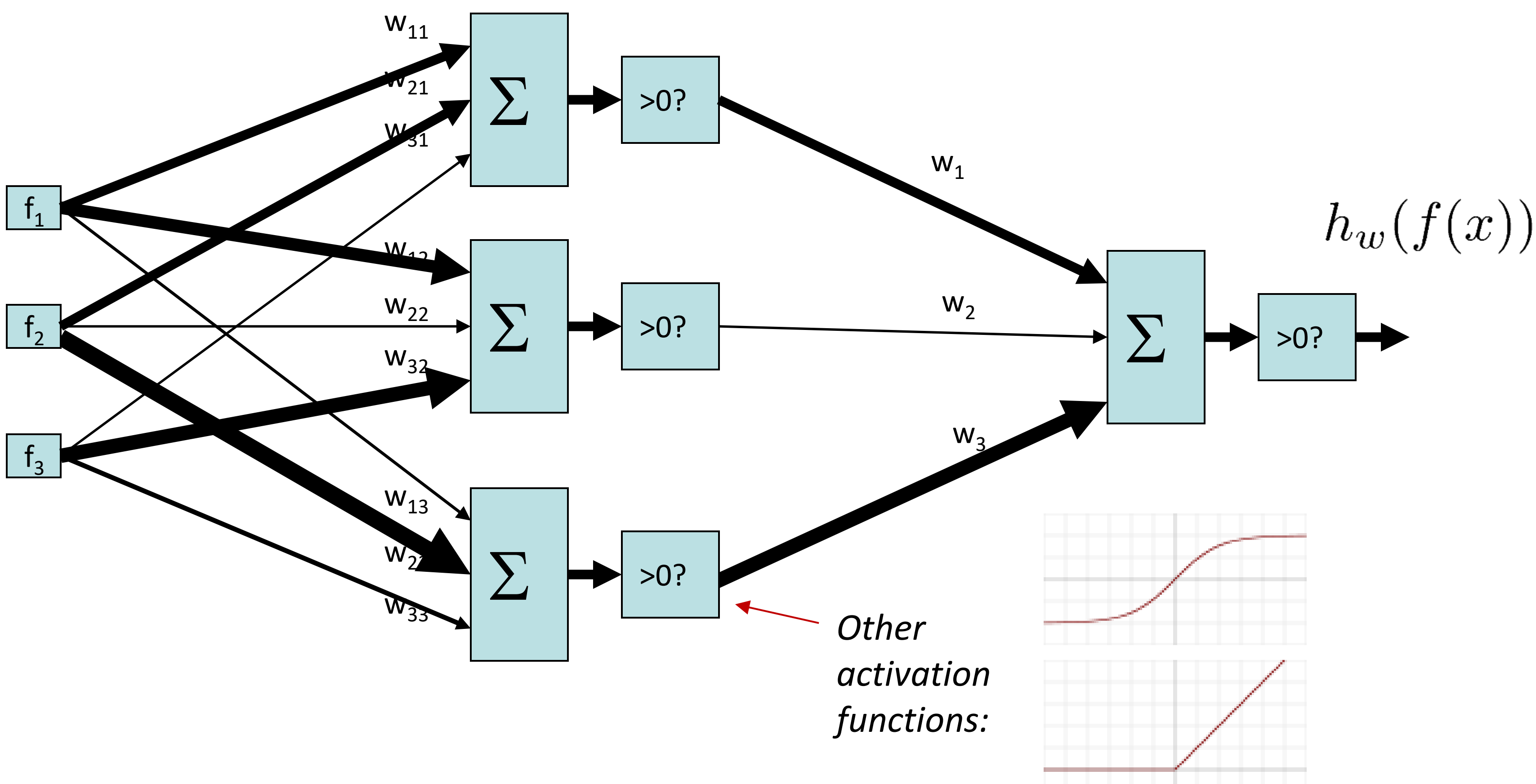


$$\sum_i w_i \cdot f_i(x) = w \cdot f(x)$$

- Output:
  - +1 if ≥0
  - -1 if <0

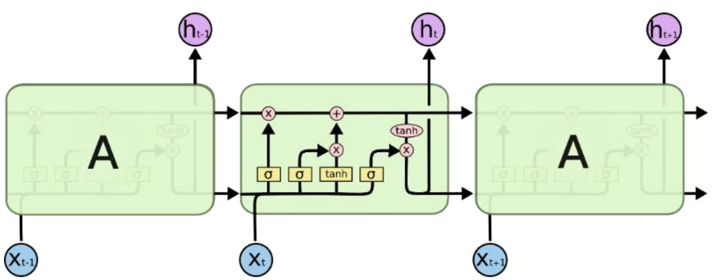


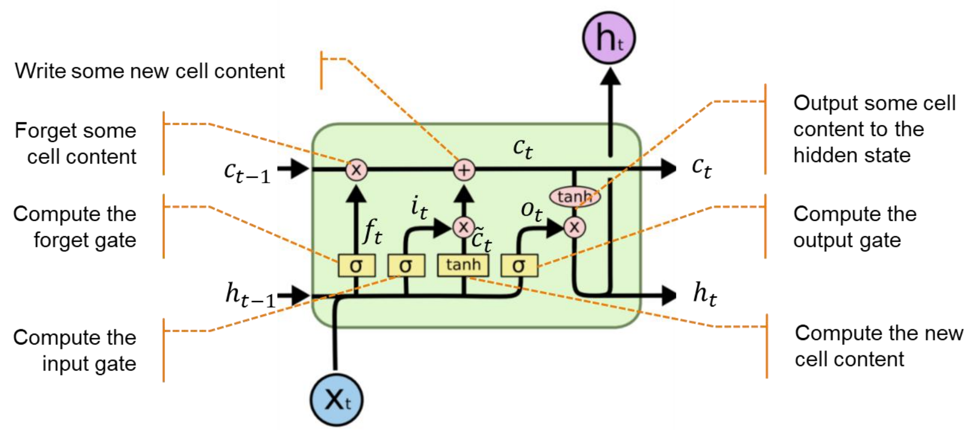
Neural networks: two-layer perceptron



Neural networks: deep learning

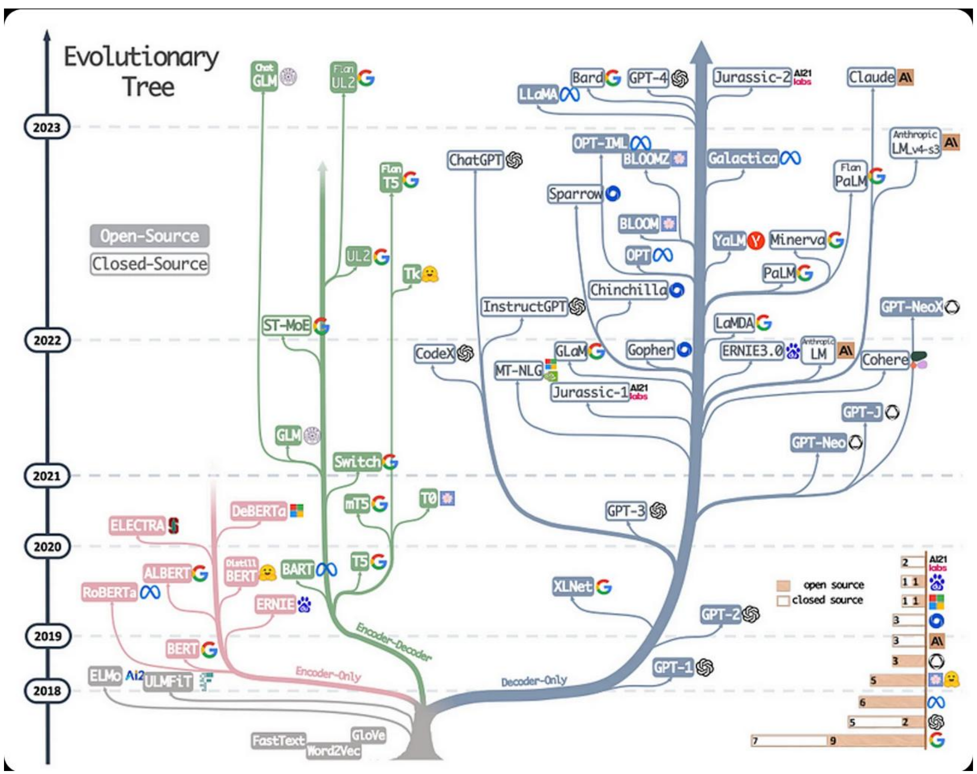
- Deep Learning
  - A large number of layers of neural networks
    - Ex. 1000 layers in ResNet
  - More complicated connections between layers
    - Ex. LSTM

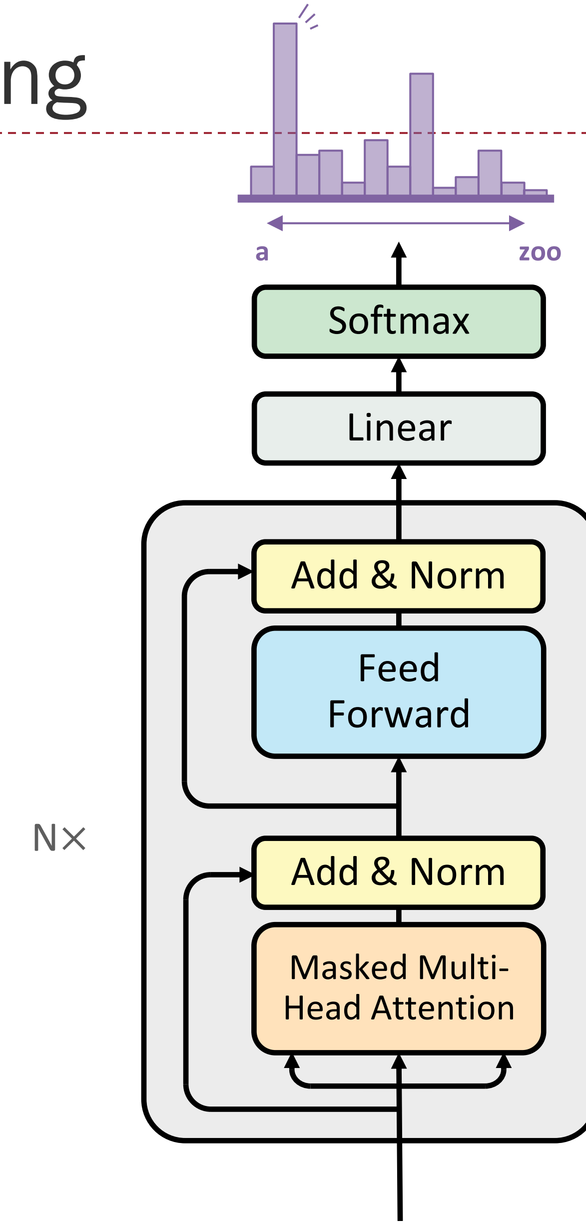




Neural networks: deep learning

- Transformer
  - The foundation of LLMs



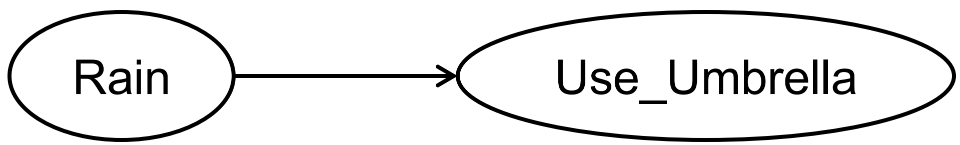


Connectionism

- History of connectionism: rose and fell for several times
  - 1940s: pioneer work, e.g., McCulloch-Pitts model
  - 1958: invention of perceptron (Rosenblatt)
  - 1969: “Perceptron” published (Minsky & Papert)
    - Publicized key issues of perceptron (e.g., XOR)
  - 1970s: AI winter
  - 1980s: revival of connectionism
    - Hopfield net, BP algorithm
    - Rumelhart & McClelland (1986): Parallel Distributed Processing
  - 1990s-2000s: overtaken in popularity by other methods
  - 2010s: rise of deep learning
    - Since ~2012: dominates CV
    - Since ~2015: dominates NLP

Statistical Approaches

- Representing knowledge with probabilistic models



	T	F
	0.2	0.8

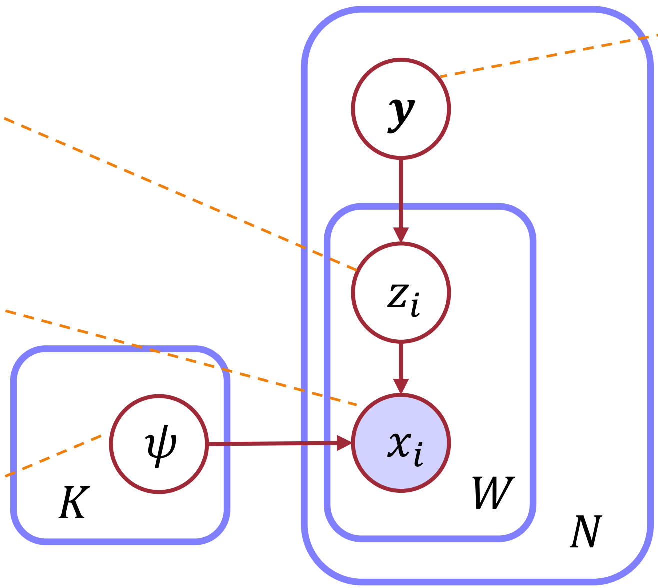
Rain	T	F
T	0.95	0.05
F	0.1	0.9

Topic Model

topic assignment for each word

word

distribution over words for each topic



distribution over topics

e.g., **[0.1, 0.3, 0.2, …]**

sports	**0.1**
arts	**0.3**
science	**0.2**
…	…

e.g., **<u>words - [*how, to, get, …*]</u>**

sports: [0.1, 0.3, 0.2, …]

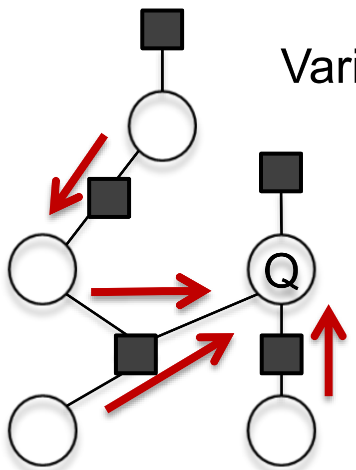
arts: [0.2, 0.3, 0.1, …]

science: [0.3, 0.4, 0.1, …]

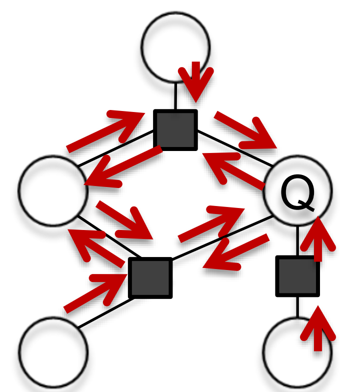
……

Statistical Approaches

- Inference and learning is done by probabilistic inference

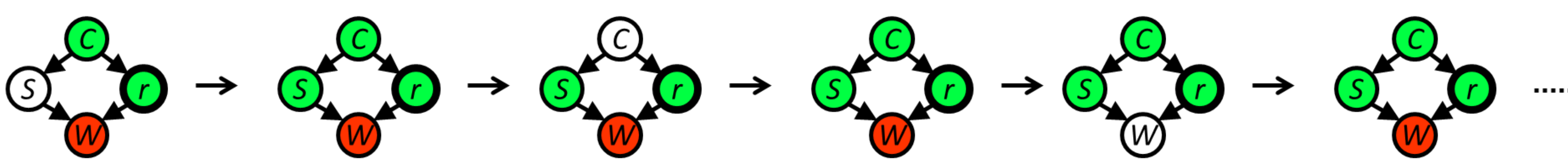


Variable Elimination



Loopy Belief Propagation

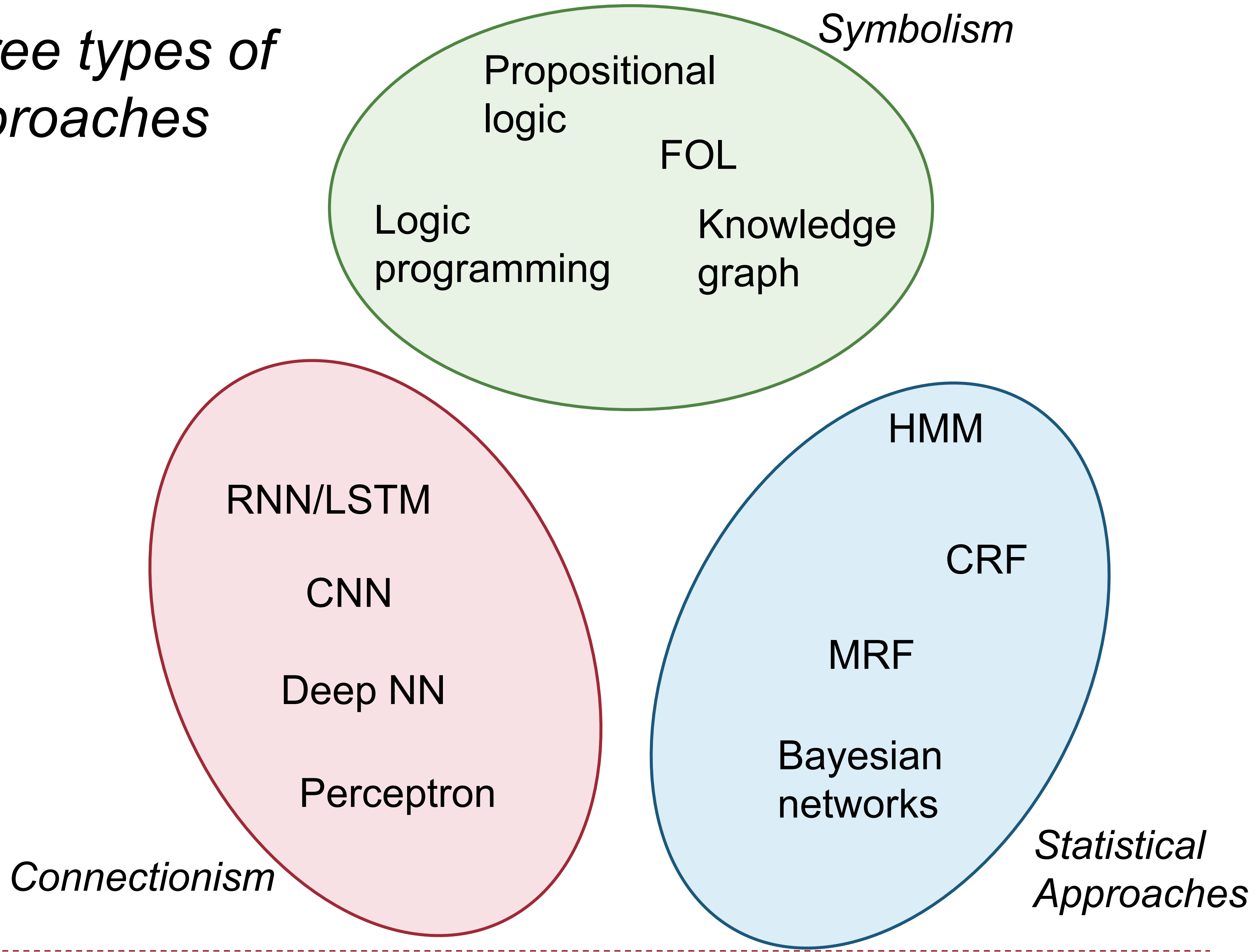
Gibbs Sampling



Statistical Approaches

- History
  - Become popular since 1990s
  - Dominant during 2000s
  - Overshadowed by deep learning in 2010s

Three types of approaches



Three types of approaches

Symbolism

Propositional

✓ Expressive, interpretable, rigorous

✗ Hard to learn, rigid

Connectionism

✓ Good performance, flexible

✗ Black-box, data-hungry, hard to incorporate knowledge or perform complex reasoning

Statistical Approaches

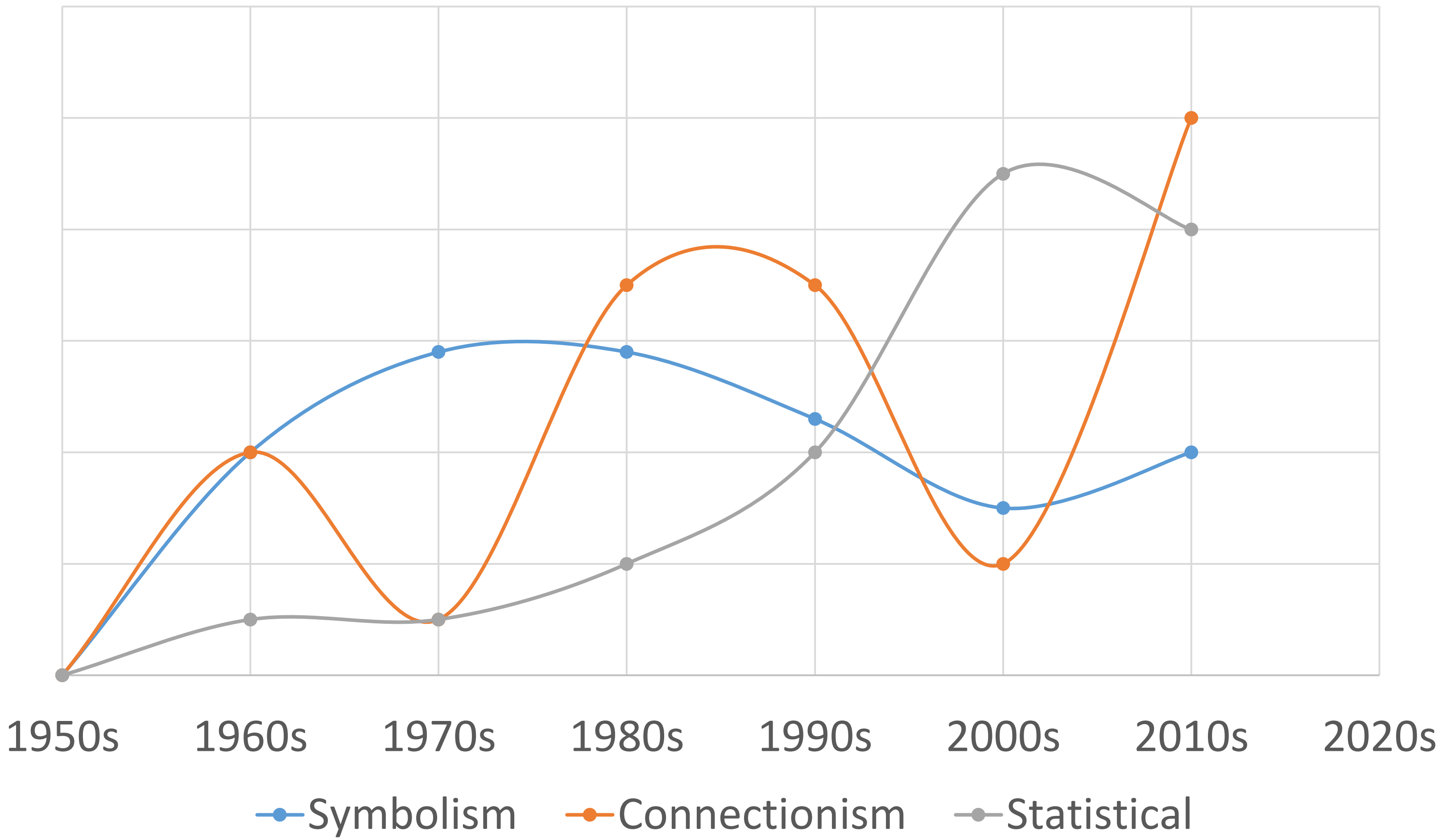
HMM

networks

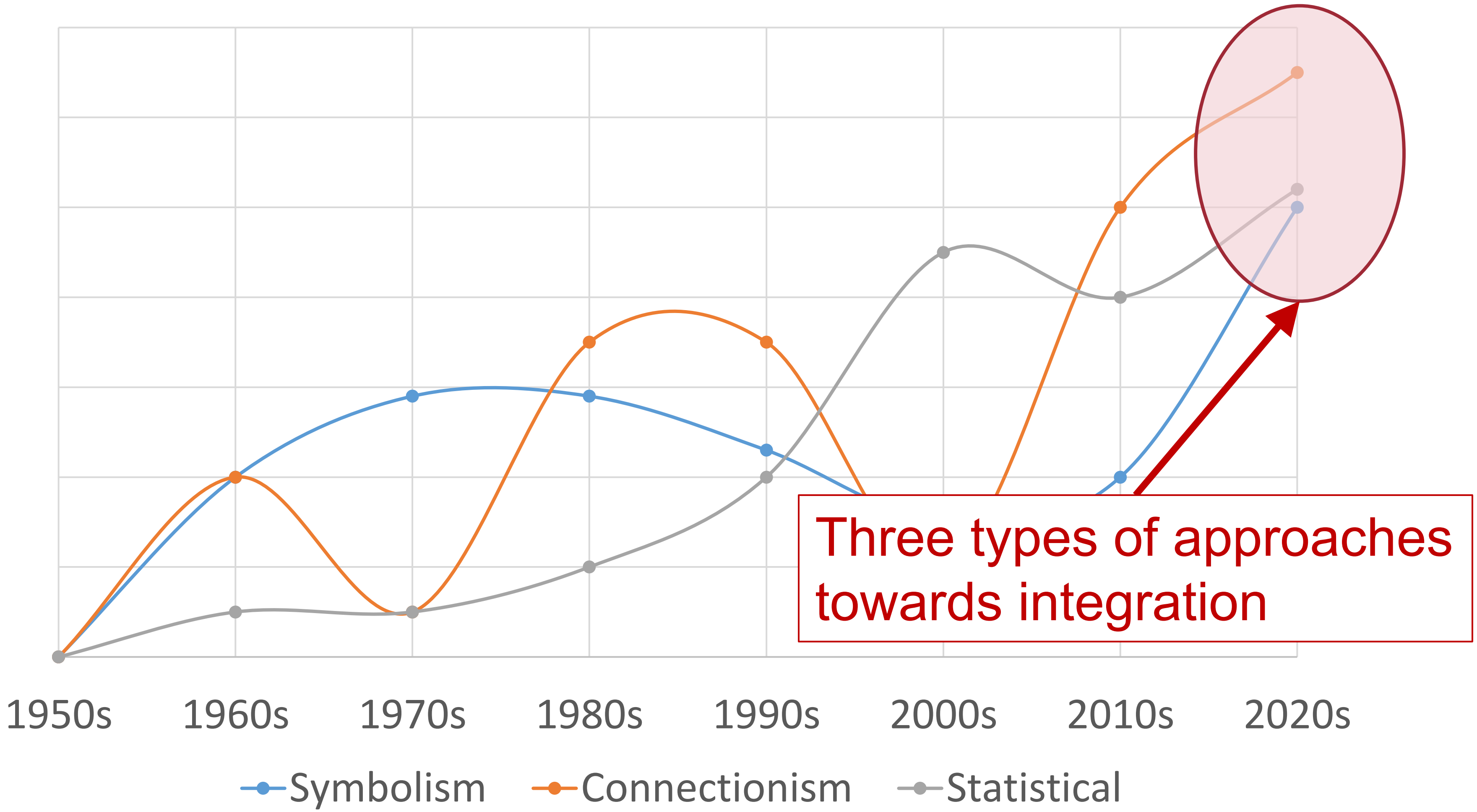
✓ Interpretable, rigorous, learnable

✗ Less expressive/ performant/flexible

Trends


1950s
1960s
1970s
1980s
1990s
2000s
2010s
2020s
Symbolism
Connectionism
Statistical

Trends



Three types of approaches towards integration
1950s
1960s
1970s
1980s
1990s
2000s
2010s
2020s
Symbolism
Connectionism
Statistical

Taking SIST AI Courses

- Undergraduate
  - CS181 Artificial Intelligence I
  - CS182 Introduction to Machine Learning
  - CS172 Computer vision I
  - CS173 Data mining
  - CS183 Introduction to Robotics
- Graduate
  - CS280 Deep Learning
  - CS282 Machine Learning
  - SI252 Reinforcement Learning
  - CS272 Computer Vision II
  - CS274A Natural Language Processing
  - CS283 Robotics
  - CS243 Introduction to Algorithmic Game Theory
  - CS284 Simultaneous Localization and Mapping
  - CS286 AI for Science and Engineering
  - …

Doing AI Research

- Learning recent developments in AI from top conferences & arXiv (not from journals!!)
  - AI: IJCAI, AAAI
    - Caution: not top in ML, NLP, CV
  - ML: NIPS, ICML, ICLR
  - NLP: ACL, EMNLP, NAACL
  - CV: CVPR, ICCV, ECCV
  - Other: KDD, AAMAS, UAI, …

- Participating in research projects…